

Death in the Desert: A Supervised GRU3 Model Predicting Temperature and Death Rate in New Mexico

Validated with Random Forest and Extreme Gradient Boosting (XGBoost)

Presented by:

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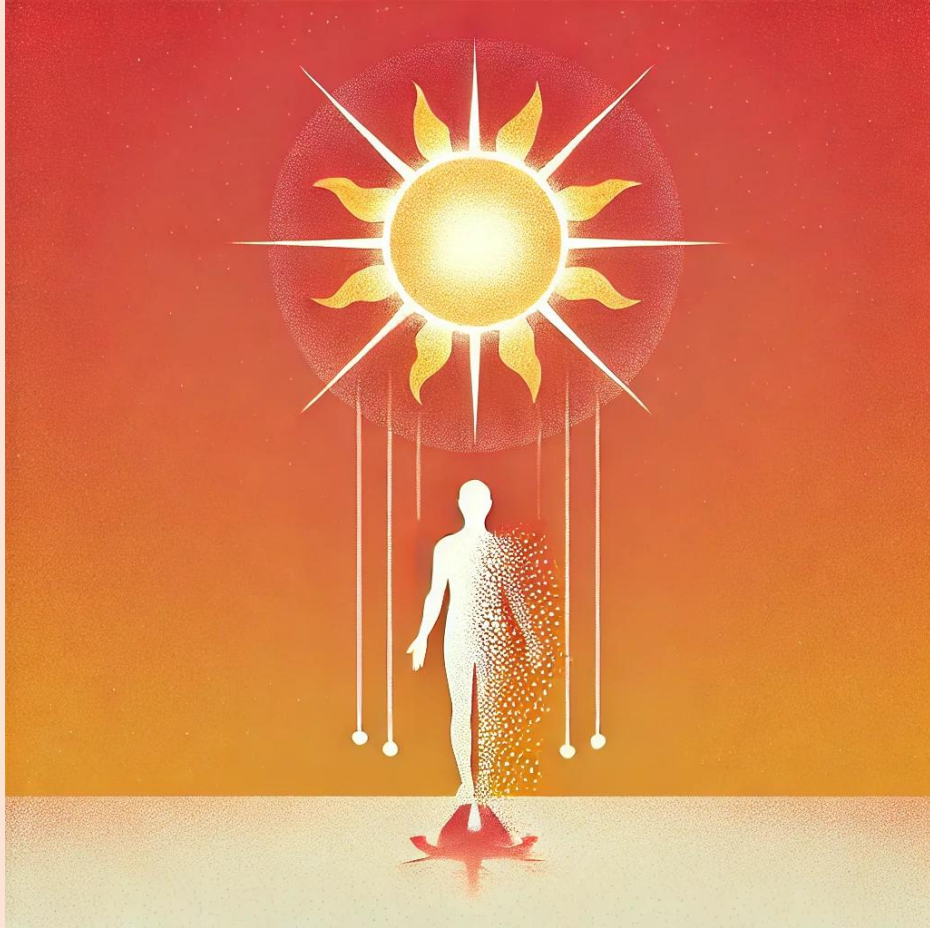


New Mexico! Land of...



Project Background and Justification

Climate and Mortality



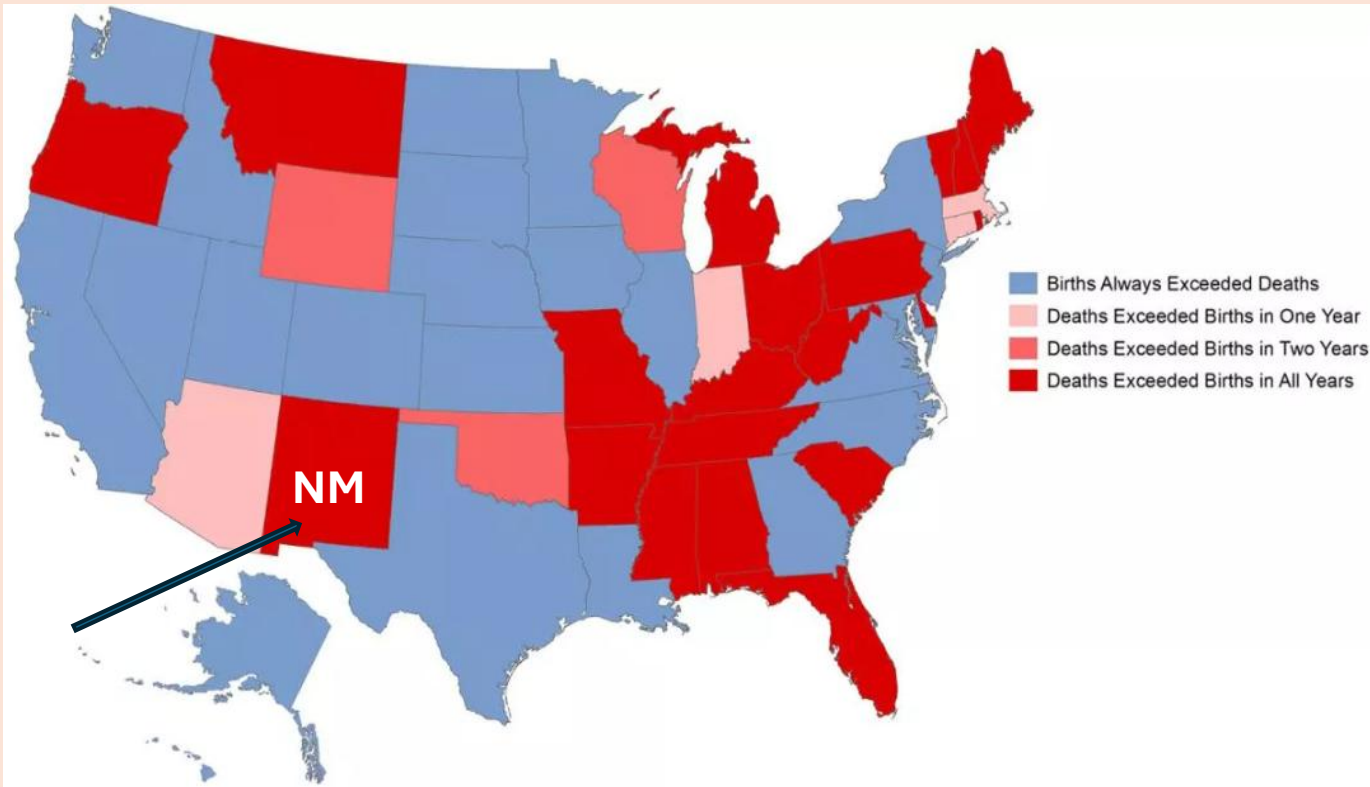
Images source: OpenAI, 2024

- The WHO (2014) identifies **rising temperatures** as one of five key factors in **increased mortality**, especially among the elderly¹.
- Studies show that without adaptation, **temperature fluctuations** could cause over 68,000 **heat-related deaths** in Europe alone by 2030².
- Thus, it has become crucial to integrate the trajectory and effects of climate change, particularly **temperature**, into **population prediction models**.

¹Guibert et al, 2024, ²Ballester et al, 2023.

Why New Mexico?

No of Years Deaths Exceed Births 2020 to 2022



Johnson, 2023

- **NM among the 20 worst performing states and is the worst in the southwest region.**

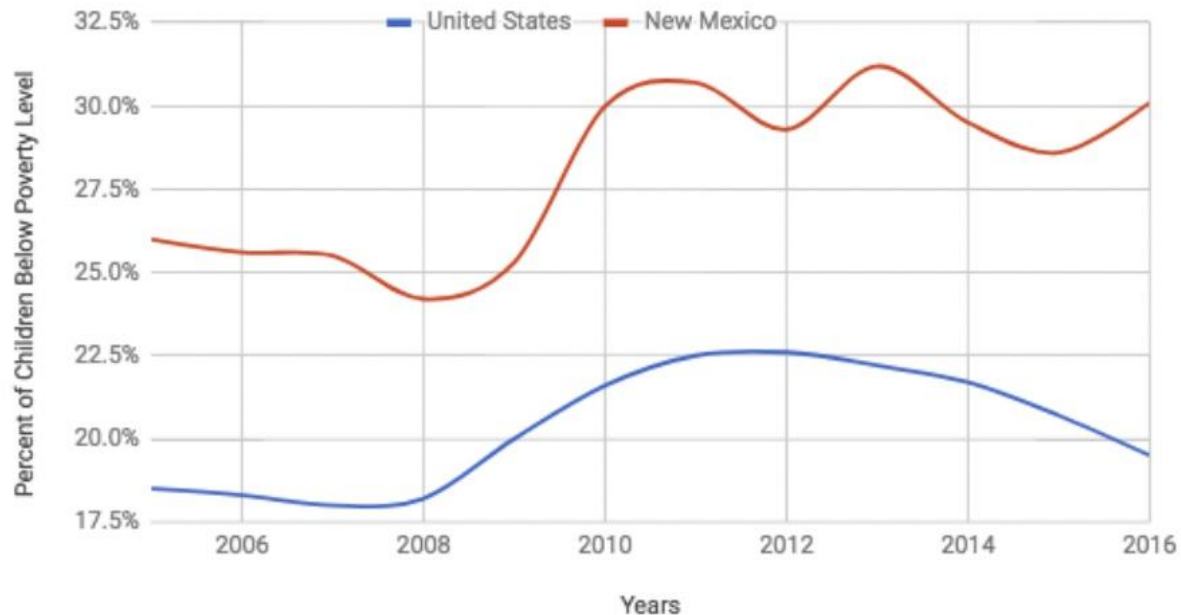
Predictor Variables

- 🌍 Minimum Temperature
- 🌍 Precipitation
- 🌍 Cooling Degree Days
- 🌍 Heating Degree Days
- 🌍 Palmer Drought Severity
- 🌍 Palmer Z-Index

More on New Mexico...



United States vs. New Mexico



Also, the land of...

- Second-fastest -warming state in the nation.
- Rapid Desertification, and
- 30% children below poverty level, 18.4% adults below poverty level.

Model Approach: New Mexico's meteorological data is used to predict and forecast *Average Temperature*, which in turn can be used to predict *Death Rate*.



Data Wrangling & Visualization

- NOAA: 9 features by month (1895-2024)
- New Mexico Center for Health Statistics (1960-2017)



Models

- Temp: GRU Type 3: Time Series Multiple Nonlinear Regression (Dense Deep Learning), Random Forest and XG Boost.
- Death Rate: Dense Deep learning model



Output

- Predict average temperature and death rate 60 years in the future

Data Wrangling



National Centers for
Environmental Information
NATIONAL OCEANIC AND ATMOSPHERIC ADMINISTRATION



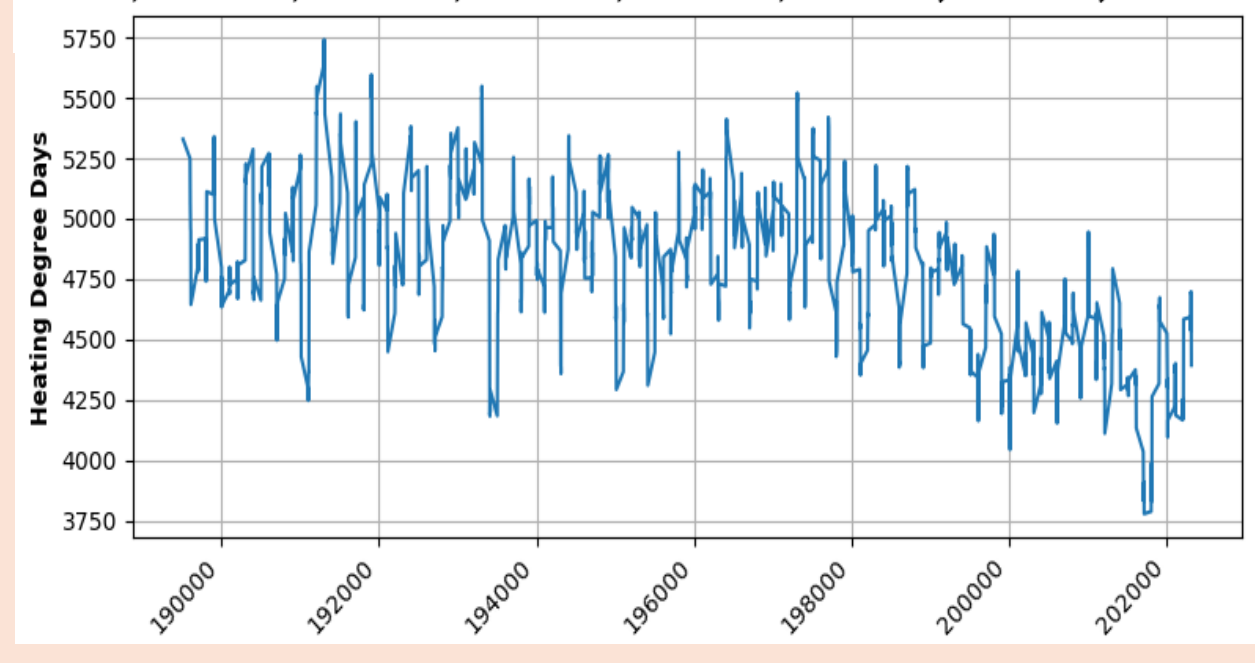
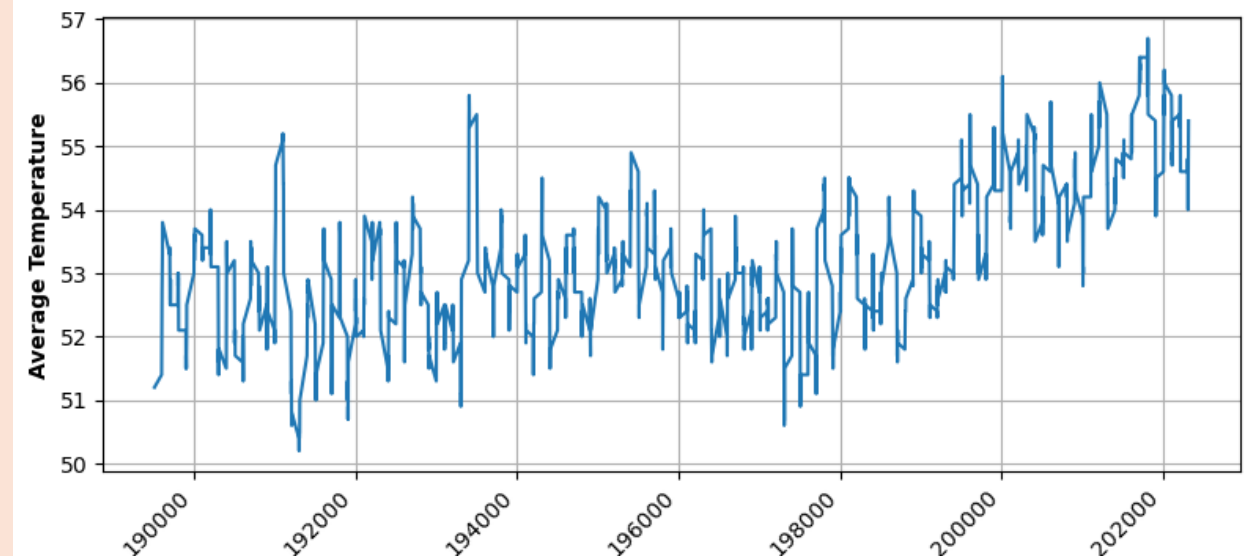
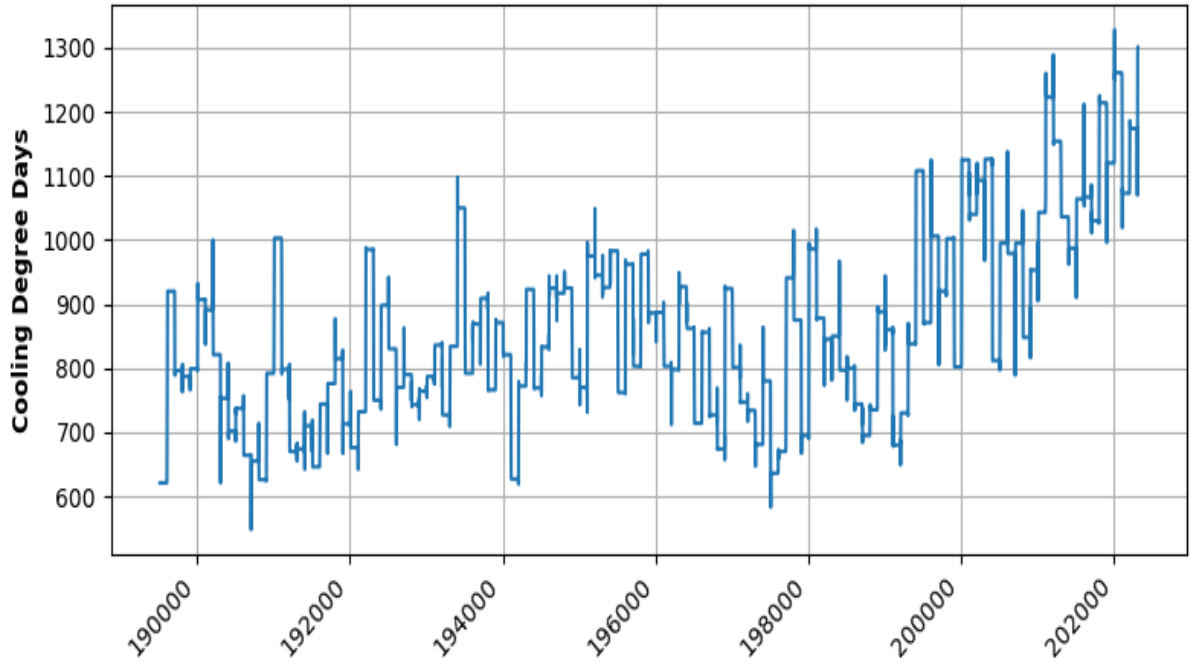
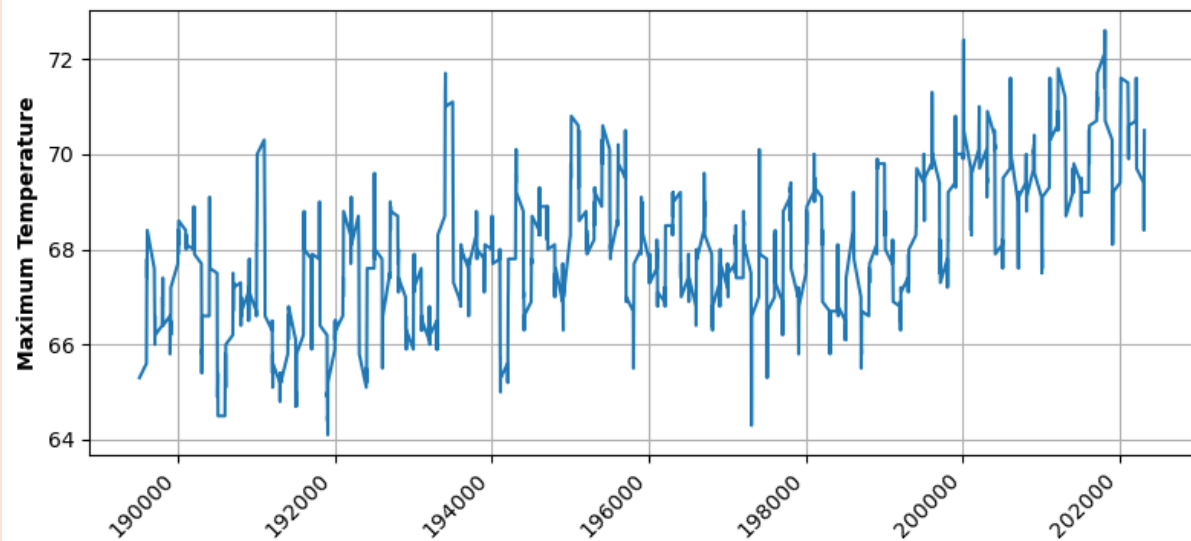
TensorFlow



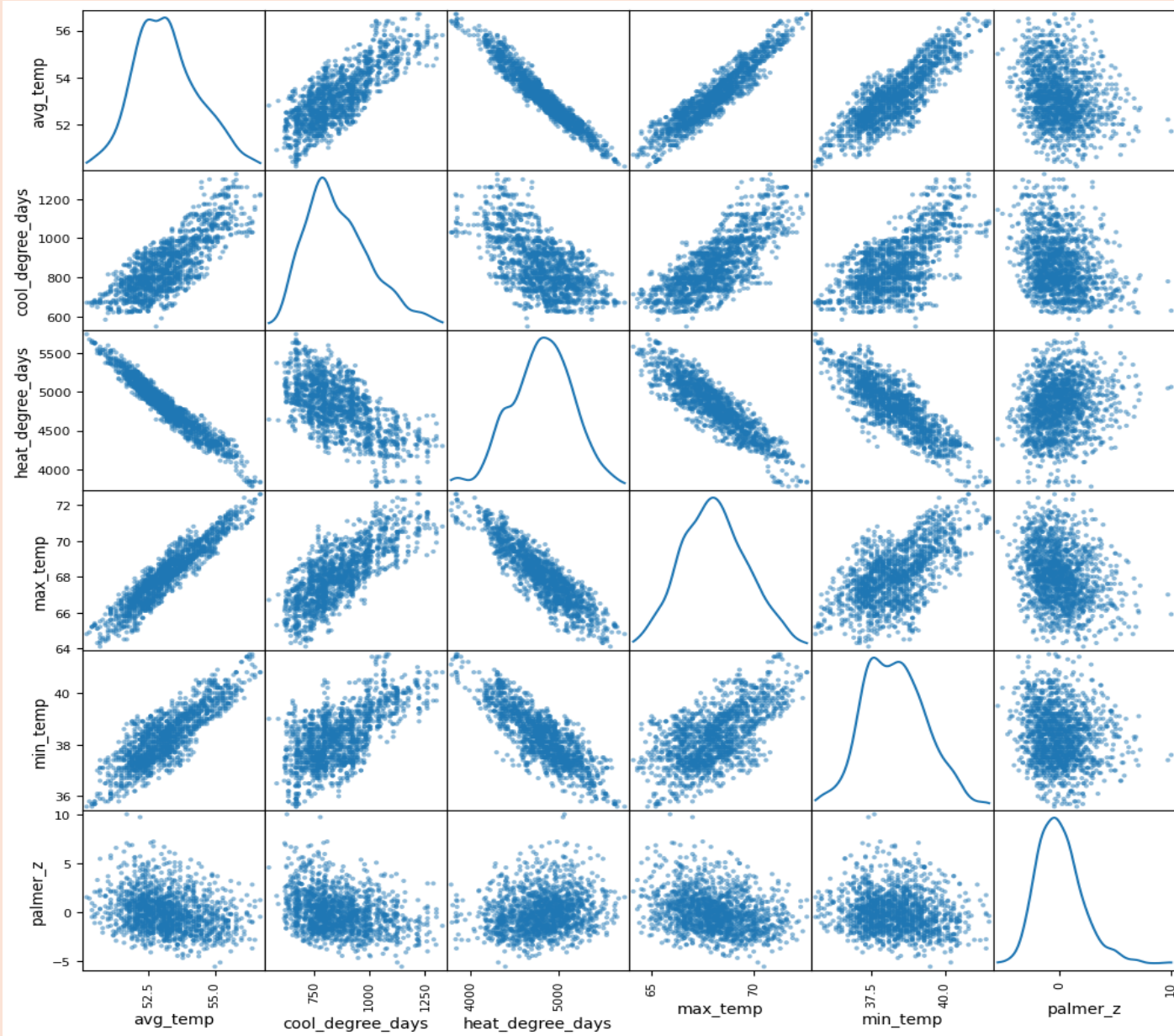
The data was sourced from the **National Oceanic and Atmospheric Administration's** (NOAA) Climate Data Online alongside the **New Mexico Department of Health**.

Little to no data pre-processing – utilized Python package pandas to append data into one large dataframe for use.

Data Visualization



Exploratory Data Analysis

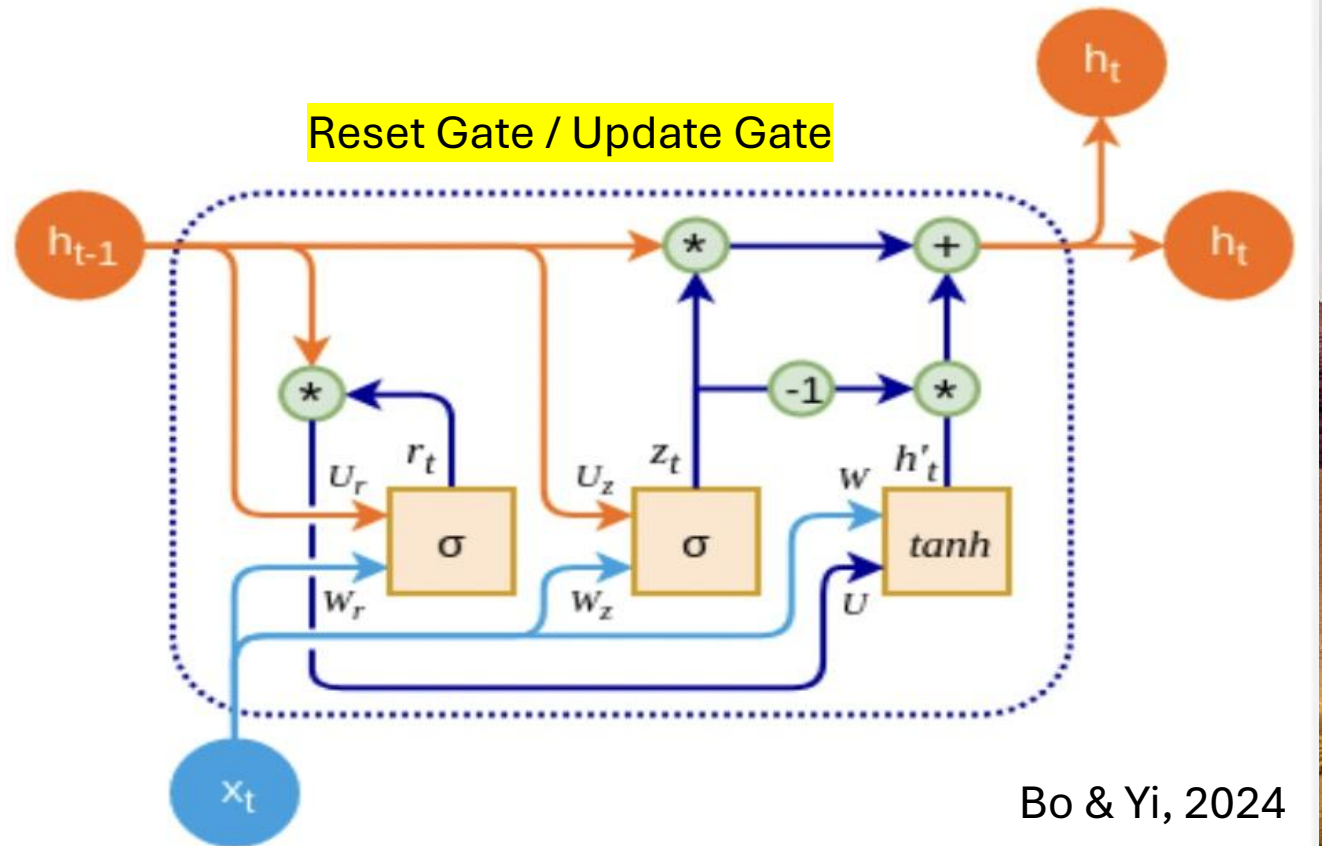


The scatter matrix clearly shows;

- **Strong positive correlation between Average Temperature, Cooling degree days, Maximum and Minimum Temperatures.**
- **Heating Degree Days exhibit a strong negative correlation with the other features.**
- **The relationship between palmer z-index and other features appears to be non-linear.**
- **No outlier observed.**

Gated Recurrent Unit (GRU)

- A GRU is a type of Recurrent Neural Network (RNN) which is a deep neural network trained on **sequential** data to produce **sequential** predictions.
- Employs **gating mechanisms** to control the flow of information throughout each time step.



GRUs are especially effective for **time-series data** – notably, for **forecasting predictions**, which is important in our case, as we want to predict average temperature and deaths over the next 60 years.

Able to retain **long-term dependencies**; absolutely necessary considering we want to understand **long-term patterns, cumulative effects** of weather, etc.

More **computationally efficient** than LSTMs, which is crucial with the amount of data we're working with.

Article | [Open access](#) | Published: 19 June 2023

Explainable sequence-to-sequence GRU neural network for pollution forecasting

[Sara Mirzavand Borujeni](#), [Leila Arras](#), [Vignesh Srinivasan](#) & [Wojciech Samek](#) 

[Scientific Reports](#) **13**, Article number: 9940 (2023) | [Cite this article](#)

2840 Accesses | **11** Citations | **1** Altmetric | [Metrics](#)

```
for param in parameters:
    print(f"PARAM", param)
    y = data_series[param].to_numpy()
    X_train, X_test, y_train, y_test = train_test_split(date_splice, y, test_size=0.2, shuffle=False, random_state=42)

    model = Sequential([
        Dense(5, activation='relu', input_shape=(X_train.shape[1],)), # input_shape=(2,)
        Dense(1) # Output layer
    ])

    model.compile(optimizer='adam', loss='mse')
    model.fit(X_train, y_train, epochs=50, batch_size=64, validation_data=(X_test, y_test), verbose=1)
```

Methodology

- Fit **GRU** models for 'avg_temp'/'deaths', our **target variables**, on all parameters
- Create **Sequential** models for all individual parameters (one input / one output)
- Use those models to **predict parameter values** for 60 years in the future
- Use predicted params for **GRU input** to predict 'avg_temp'/'deaths'.

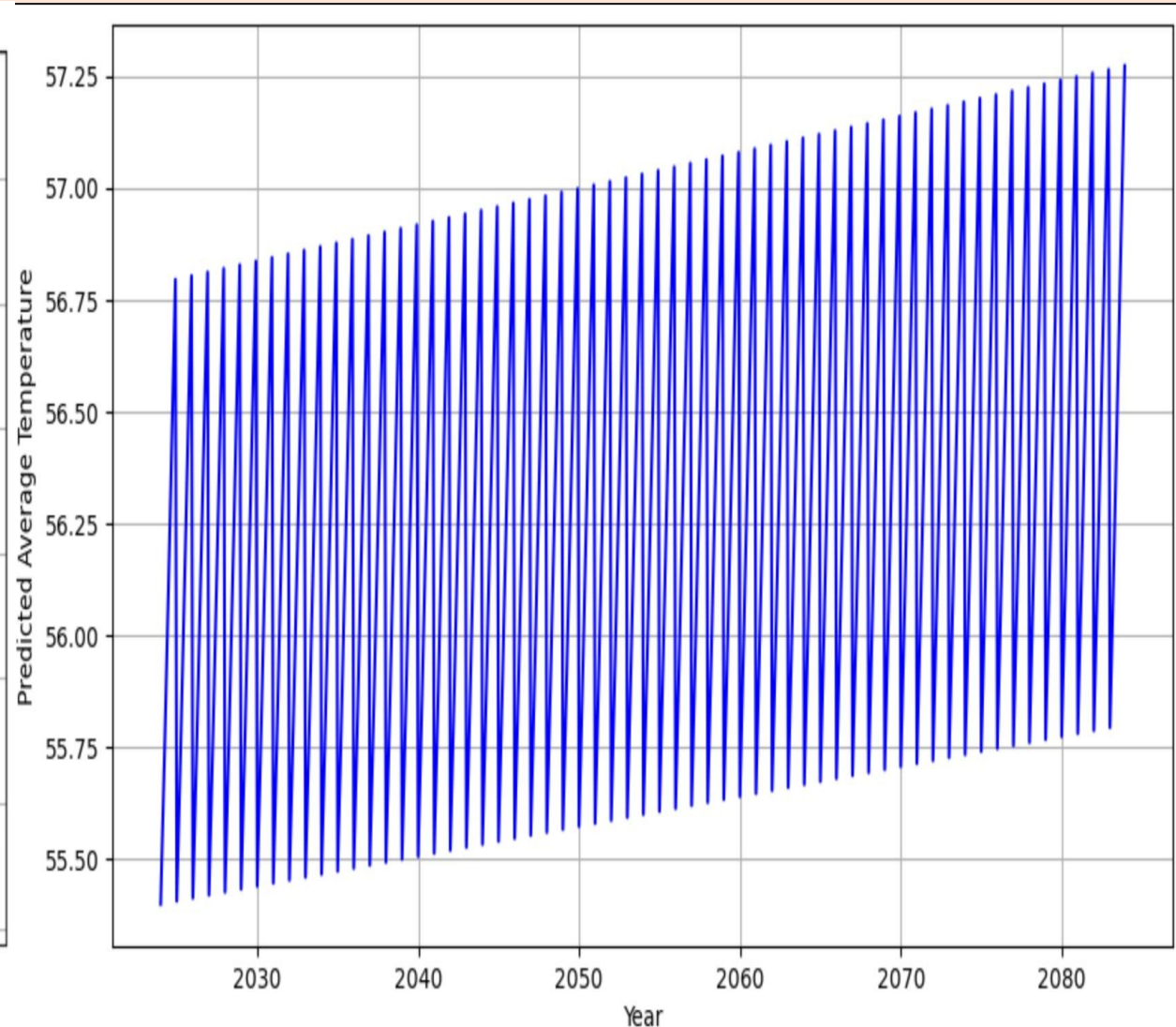
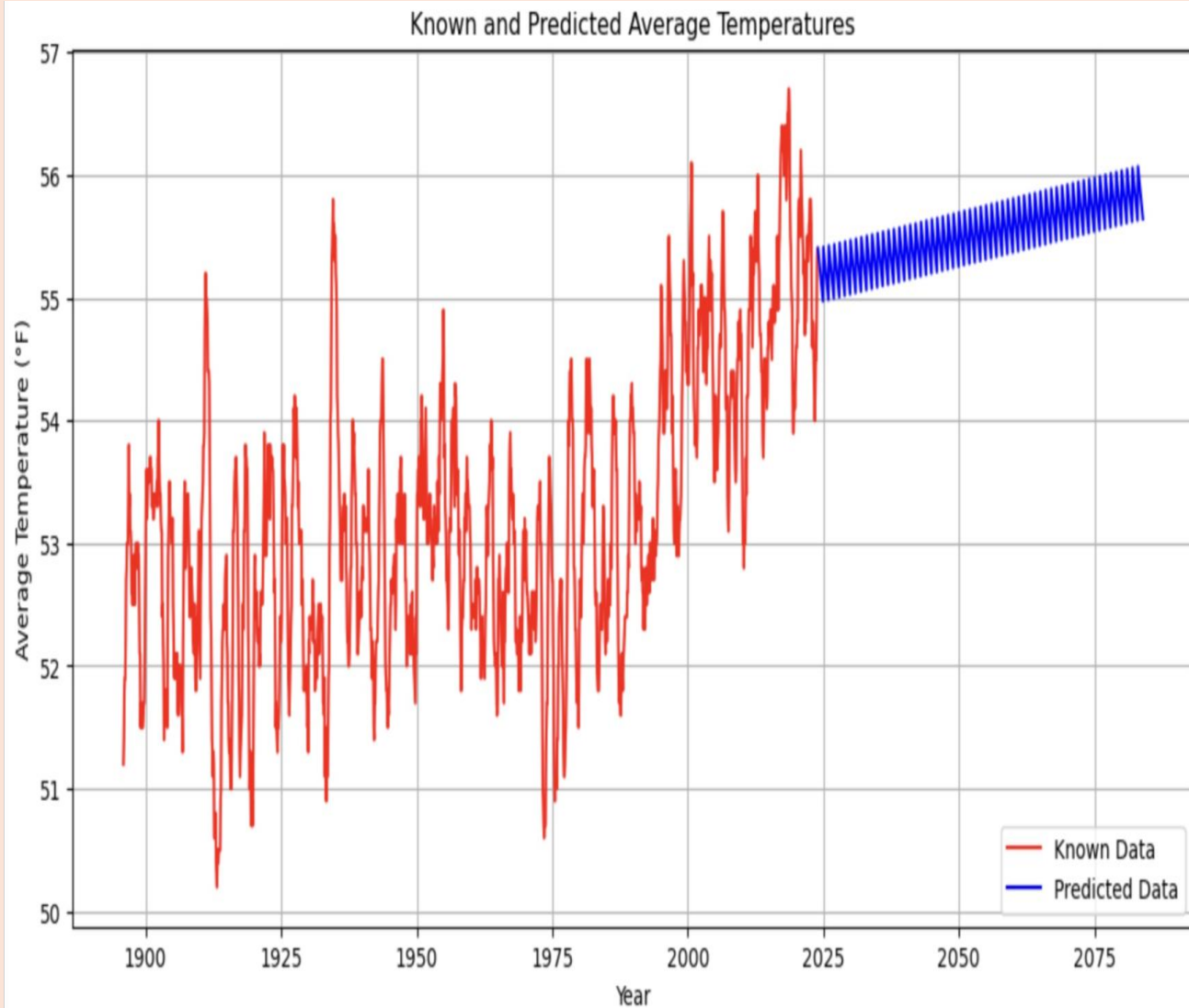
Building the GRU

```
# Train the GRU model
model_gru = Sequential(
    [
        GRU(20, activation='relu', input_shape=(x.shape[1], x.shape[2])),
        Dense(1)
    ]
)

model_gru.compile(optimizer='adam', loss='mse')
model_gru.fit(x, avg_temp, epochs=300, batch_size=1, verbose=0)
```

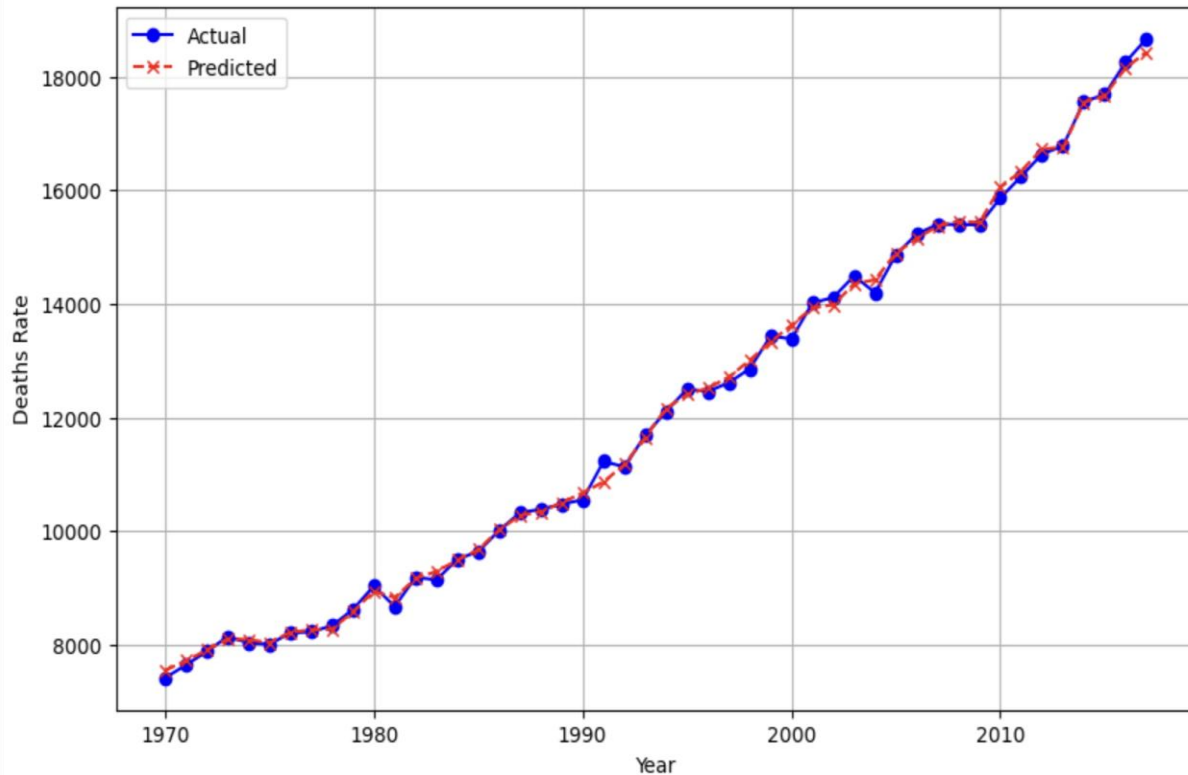


Average Temperature

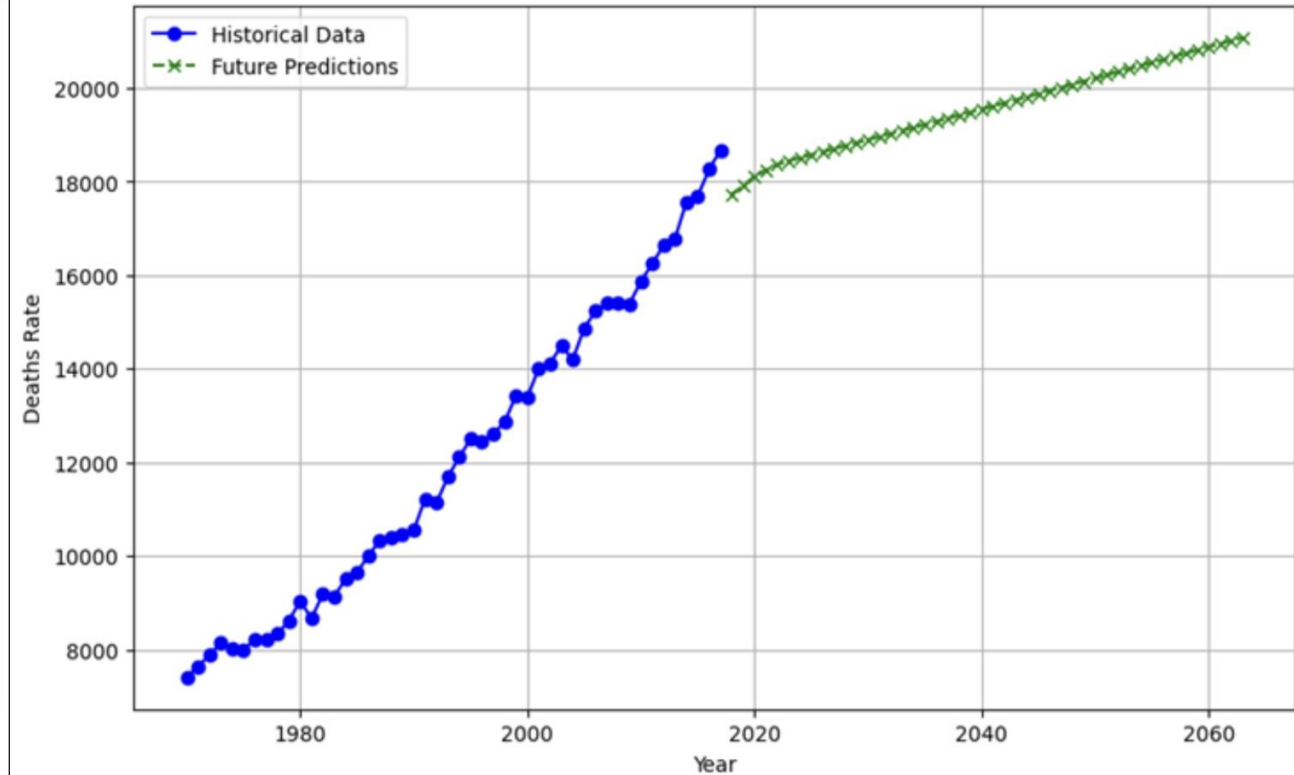


Death Rate Result

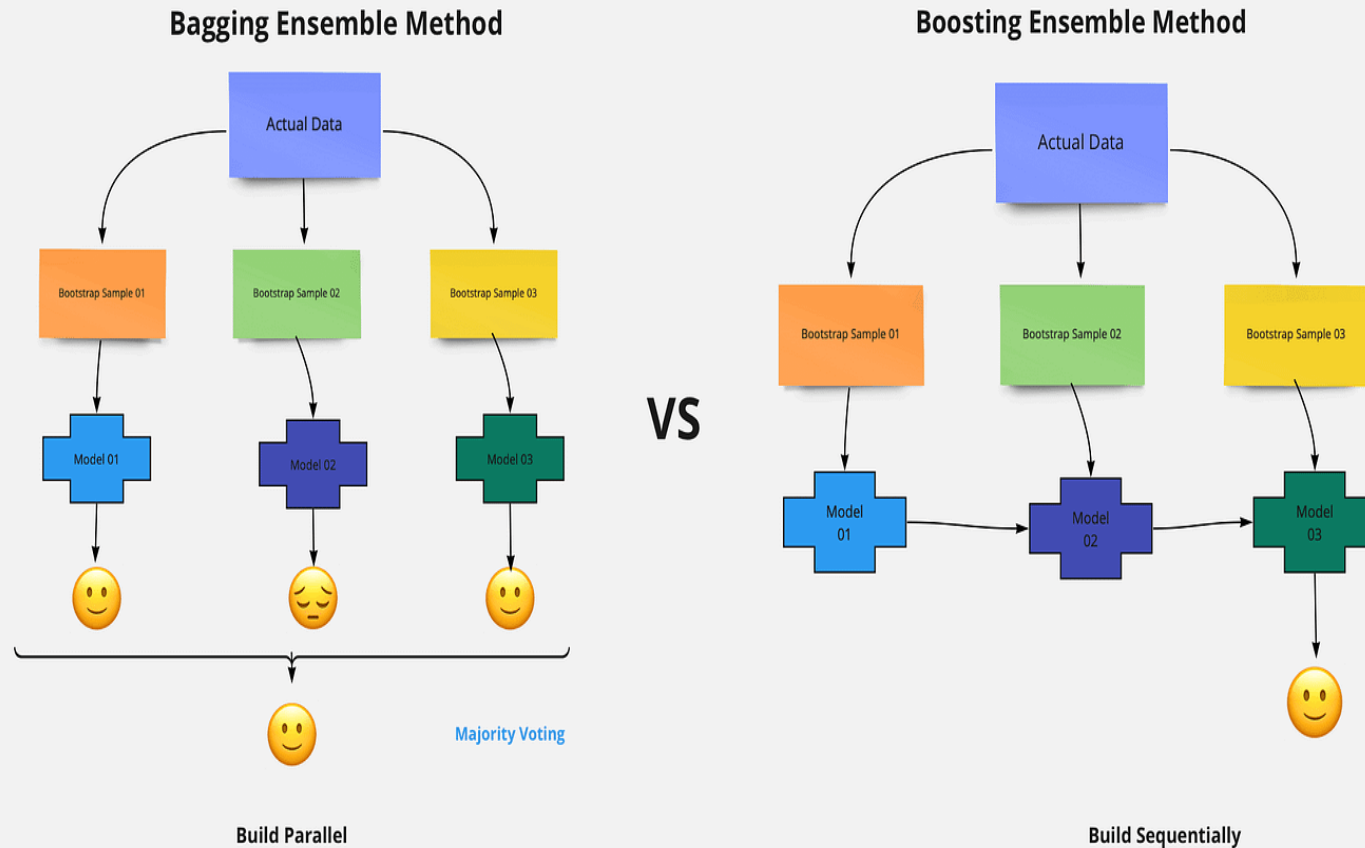
Actual vs. Predicted Values



Historical Data with Immediate Future Predictions



Random Forest Vs. XGBoost



dataaspirant.com

1. Algorithm Type

- Random Forest: Bagging (Parallel tree construction).
- XGBoost: Boosting (Sequential tree construction).

2. Speed and Performance:

- Random Forest: Faster training due to parallelism.
- XGBoost: Slower but optimized for performance.

3. Overfitting Control:

- Random Forest: Less prone to overfitting due to averaging.
- XGBoost: Includes L1/L2 regularization to control overfitting.

Conclusion

Identified the risk of New Mexico's aging population with global warming



Designed a predictive algorithm for average temperatures of 60 years



Dense layer for death rate



Show dependency of death rate in correlation with avg temp

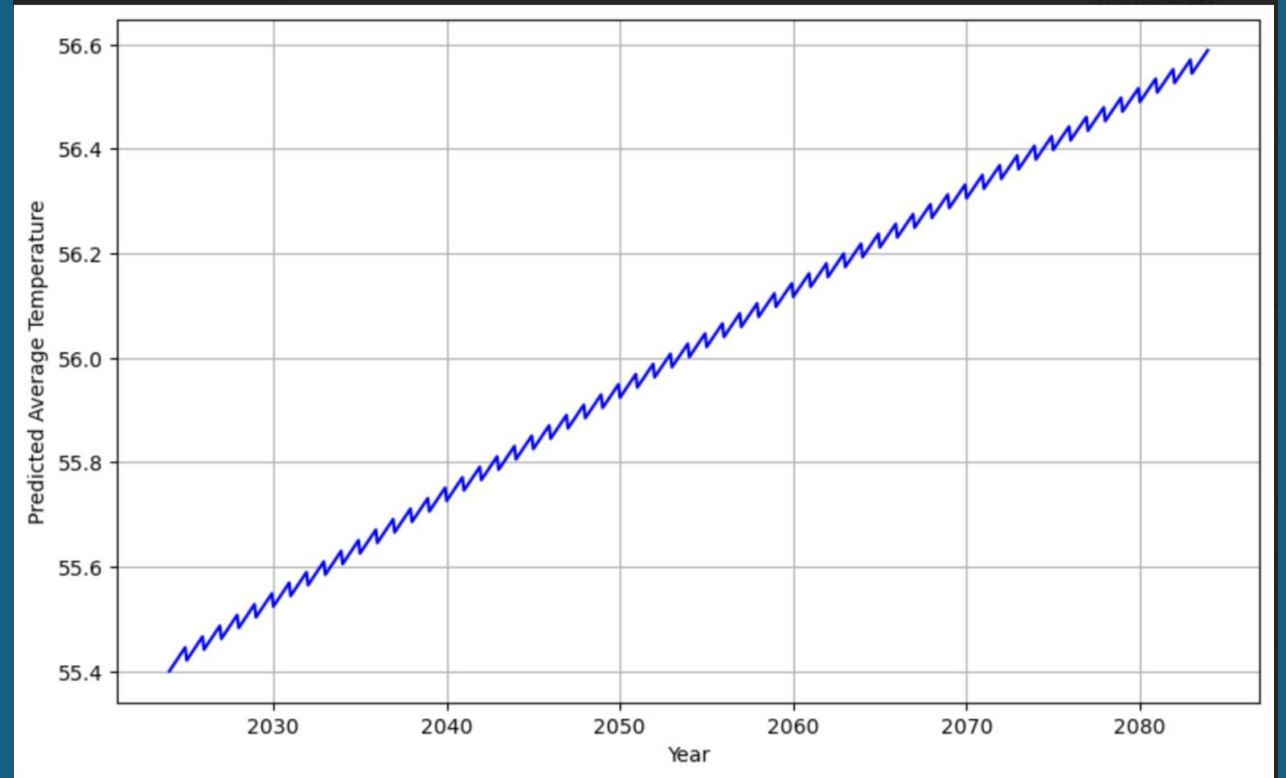
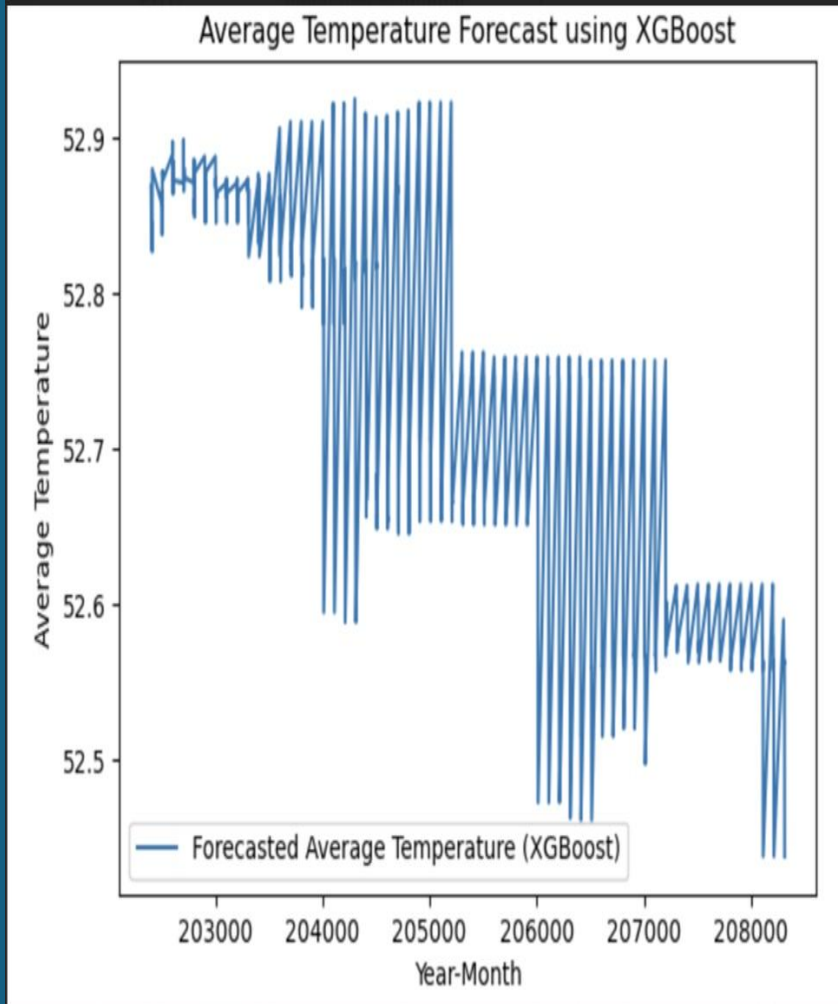


Inform policy and urban development to mitigate

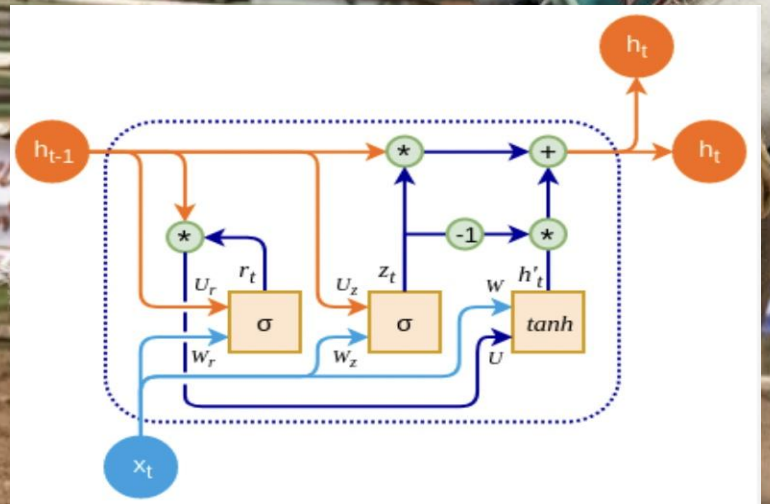
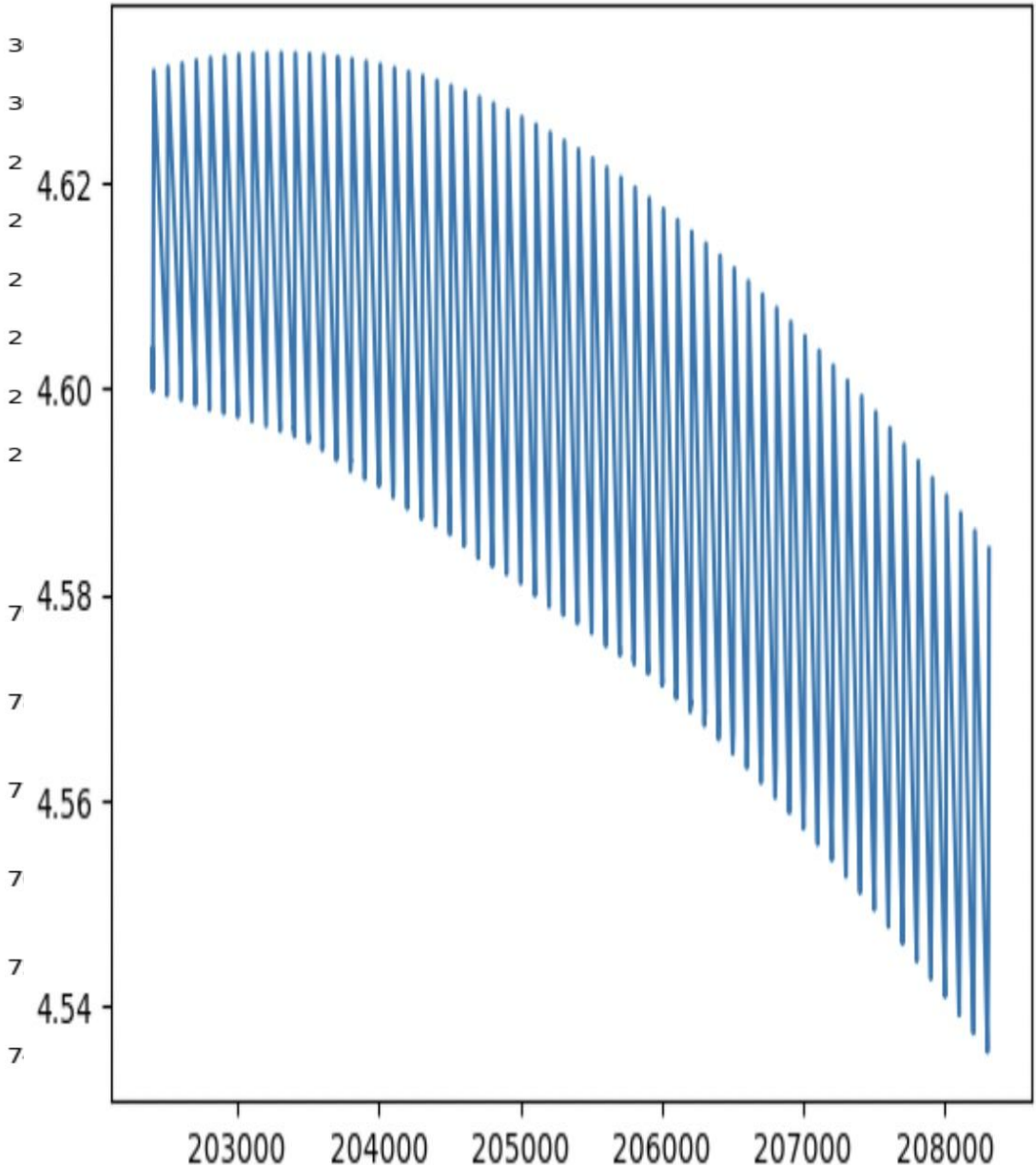


Thank you for listening! Questions?

Q/A Slide: So, you've asked about the Random Tree and the XGBBoosting!



Q/A: So, you've asked about trials and Tribulations



Building Our Virtual Environment

```
from tensorflow import keras
from keras.models import Sequential
from keras.layers import Dense, GRU
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error
```

Libraries used; GRU from TensorFlow V 2.16.1

```
1 # Shell script to install the required packages for the project
2 #!/bin/bash
3
4 # Ensure conda is installed and sourced
5
6 conda create --name ml_new_mexico python tensorflow
7 conda activate ml_new_mexico
8
9 pip install numpy pandas matplotlib scikit-learn keras ipykernel jupyter
10
11 python -m ipykernel install --user --name=ml_new_mexico
```

Bash Script



TensorFlow

matplotlib



NumPy



pandas



Keras

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